

ORF307: Optimization

Precept: #5

Agenda

- Converting LPs
- Steel company operations
- The moment problem (Optimization helps Probability Theory)
- Chebyshev center (maximum inscribed ball)

Reminders and Announcements

- Always check Python notebooks in Companion (in directories `lectures/` and `precepts/`)
- Write down the deadline of the current HW:
- Other announcements on blackboard.

Converting LPs

Let $A \in \mathbb{R}^{m \times n}$, $x \in \mathbb{R}^n$, $b \in \mathbb{R}^m$, and $k \in \mathbb{R}$.

We would like to convert the constrained optimization problem

$$\begin{array}{ll} \min_x & \|Ax - b\|_1 \\ \text{s.t.} & \|x\|_\infty \leq k \end{array} \quad (1)$$

into the following two formulations:

- Formulation 1:

$$\begin{array}{ll} \min_{\tilde{x}} & \tilde{c}^T \tilde{x} \\ \text{s.t.} & \tilde{A} \tilde{x} = \tilde{b} \\ & \tilde{C} \tilde{x} = \tilde{d} \end{array}$$

- Formulation 2:

$$\begin{array}{ll} \min_x & c^T x \\ \text{s.t.} & Ax = b \\ & x \geq 0 \end{array}$$

Solution:

Let us start with Formulation 1.

- Recall that the L_∞ norm is defined as

$$\|x\|_\infty = \max_i \{|x_i|\}.$$

Then the constraint $\|x\|_\infty \leq k$ can be equivalently expressed by the $2n$ inequalities

$$x_i \leq k \quad \text{and} \quad -x_i \leq k, \quad \forall i = 1, 2, \dots, n$$

- Let $s \in \mathbb{R}^m$ be a proxy of $|Ax - b|$. Then

$$a_i^T x - b_i \leq s_i \quad \text{and} \quad -a_i^T x + b_i \leq s_i$$

or equivalently (and more compactly),

$$Ax - b \leq s \quad \text{and} \quad -Ax + b \leq s$$

Remark: This is a standard trick that replaces each absolute value $|z_i|$ with an auxiliary variable s_i and a pair of linear constraints: $z_i \leq s_i$ and $-z_i \leq s_i$.

- Then the original optimization problem (1) can be equivalently written as

$$\begin{aligned} \min_{x,s} \quad & \mathbf{1}^T s \\ \text{s.t.} \quad & x \leq \mathbf{1}k \\ & -x \leq \mathbf{1}k \end{aligned} \tag{2}$$

$$\begin{aligned} & Ax - b \leq s \\ & -Ax + b \leq s \end{aligned} \tag{3}$$

- Now if we define the following vectors and matrices:

$$\tilde{c} = \begin{pmatrix} 0 \\ \mathbf{1} \end{pmatrix}, \quad \tilde{A} = \begin{bmatrix} I & 0 \\ -I & 0 \\ A & -I \\ -A & -I \end{bmatrix}, \quad \tilde{x} = \begin{pmatrix} x \\ s \end{pmatrix}, \quad \tilde{b} = \begin{pmatrix} \mathbf{1}k \\ \mathbf{1}k \\ b \\ -b \end{pmatrix},$$

we can convert the original problem (2) into Formulation 1, where $\tilde{C} = 0$ and $\tilde{d} = 0$.

To come up with Formulation 2 you have to convert the inequalities in (2) into equalities by adding slack variables and then defined the appropriate vectors and matrices. It is left as exercise.

Remark:

Table 1: Data for the steel company.

	Production rate (tons/hr)	Revenue (\$/ton)	upper bounds (tons)
Bands	200	25	6000
Coils	140	30	4000

Steel company operations

- A steel company can produce bands and coils
- The goal is to maximize its revenue
- The available data is shown in the following table
- There are 40 hours of production time this week
- Decide how many tons of bands and coils should be produced to maximize revenue.

Solution:

We will define the decision variables, objective and constraints:

- Decision variables
 - b : the number of hours allocated to produce bands
 - c : the number of hours allocated to produce coils
- Objective function: maximize total revenue. Note that the company's hourly revenue from bands is 200×25 and 140×30 , respectively. Hence we obtain the objective:

$$\max 200 \times 25b + 140 \times 30c \implies \max 5000b + 4200c$$

- Constraints:

$$\begin{aligned}
 b + c &\leq 40 \\
 200b &\leq 6000 \rightarrow b \leq 30 \\
 140c &\leq 4000 \rightarrow c \leq 28.6 \\
 b, c &\geq 0
 \end{aligned} \tag{4}$$

We see that it is better to produce as many bands as possible since the coefficient of b is larger than the coefficient of c in the objective function ($5000 > 4200$).

Verify that the optimal solution is $b^* = 30$ and $c^* = 10$.

Moment problem

This problem shows how to use linear programming to find the maximum and minimum possible values of a random variable's moment, given some known moments. This is useful for bounding quantities when we have incomplete information about a probability distribution. The definition of the moments of a random variable is discussed in "ORF309-Probability and Stochastic Systems".

- Suppose Z is a discrete random variables taking values in the set $\{0, 1, \dots, k\}$ with probabilities $p_0, p_1, \dots, p_k$
- We know the values of the 1st and 2nd, that is $\mathbb{E}[Z] = m_1$ and $\mathbb{E}[Z^2] = m_2$.

We note that these moments can be defined in terms of the probabilities p_i as follows:

$$\mathbb{E}[Z] = \sum_{i=0}^k p_i i \quad \text{and} \quad \mathbb{E}[Z^2] = \sum_{i=0}^k p_i i^2$$

- We would like to obtain upper and lower bounds on the 4th moment defined as

$$\mathbb{E}[Z^4] = \sum_{i=0}^k p_i i^4$$

- Show how LPs can be used to approach this problem.

Solution:

- We define as decisions variables the probabilities $p_0, p_1, \dots, p_k$
- The objective is the 4th moment. When we seek a lower bound we minimize the 4th moment, and when we seek an upper bound we maximize the 4th moment. Hence we have the objectives:

$$LB = \min_p \sum_{i=0}^k p_i i^4 \quad \text{and} \quad UB = \max_p \sum_{i=0}^k p_i i^4$$

- The constraints are common in the LB problem and the UB problem and defined as follows:
 - $\sum_{i=0}^k p_i i = \mathbb{E}[Z]$ (it enforces that the first moment (the mean) of the distribution of Z matches the known value $\mathbb{E}[Z] = m_1$)
 - $\sum_{i=0}^k p_i i^2 = \mathbb{E}[Z^2]$ (similarly as in the previous constraints, but here we have the second moment).
 - $\mathbf{1}^T p = 1$ (probabilities must add up to 1)
 - $p \geq 0$ (probabilities are always nonnegative)

Hence the two optimization problems are

- Lower bound

$$\begin{aligned}
LB = \min_p \quad & \sum_{i=0}^k p_i i^4 \\
s.t. \quad & \sum_{i=0}^k p_i i = \mathbb{E}[Z] \\
& \sum_{i=0}^k p_i i^2 = \mathbb{E}[Z^2] \\
& \mathbf{1}^T p = 1 \\
& p \geq 0
\end{aligned}$$

- Upper bound

$$\begin{aligned}
UB = \max_p \quad & \sum_{i=0}^k p_i i^4 \\
s.t. \quad & \sum_{i=0}^k p_i i = \mathbb{E}[Z] \\
& \sum_{i=0}^k p_i i^2 = \mathbb{E}[Z^2] \\
& \mathbf{1}^T p = 1 \\
& p \geq 0
\end{aligned}$$

Chebyshev center

- Consider a set P described by linear inequality constraints

$$P = \{x \in \mathbb{R}^n \mid a_i^T x \leq b_i, i = 1, 2, \dots, m\}$$

Such a set is called a polyhedron.

- **Goal 1:** find a ball with the largest possible radius that is entirely contained within the polyhedron P .
- **Goal 2:** Provide an LP formulation of this problem.

Solution:

Let us consider the plot described in Figure 1. The polyhedron consists of five affine constraints.

We are looking for the center of the circle and the radius of the circle. Therefore, the **decision variables** are $c \in \mathbb{R}^2$ and $r \in \mathbb{R}$ with $r \geq 0$.

Now let us focus on the i -th constraint, $a_i^T x \leq b_i$.

- The distance from a point c (in this case, the center of the circle) to the hyperplane $a_i^T x \leq b_i$ is defined by

$$\frac{|a_i^T c - b_i|}{\|a_i\|_2}$$

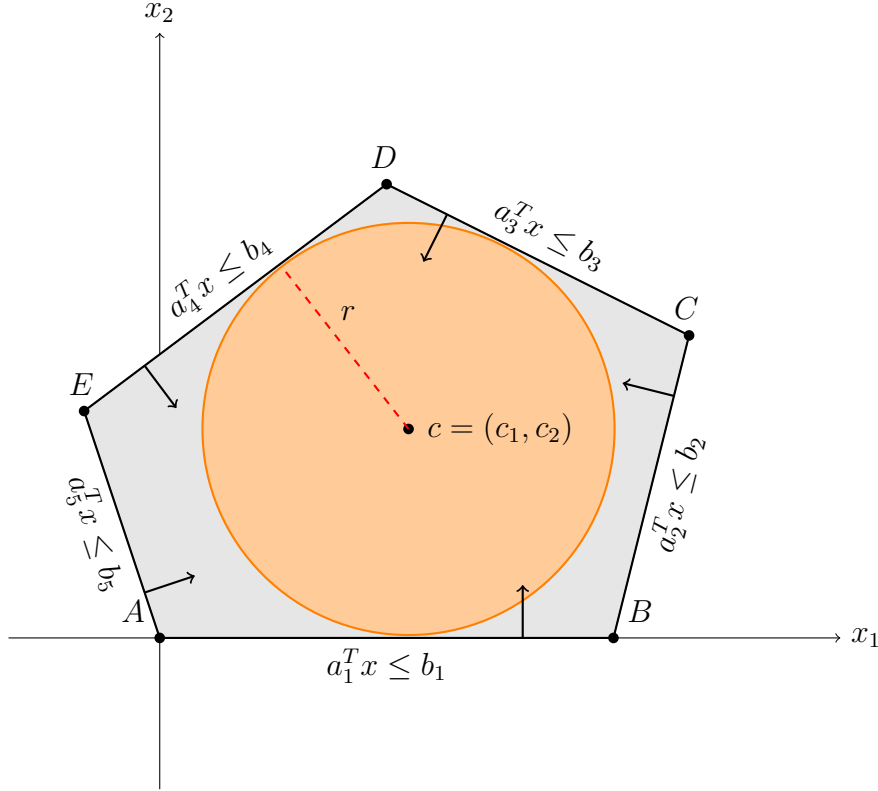


Figure 1: An example of the maximum inscribed ball within a polyhedron in 2D.

- Since the point c is within the feasible region, we must have $a_i^T c - b_i \leq 0$ and the distance becomes:

$$\frac{-a_i^T x + b_i}{\|a_i\|_2}$$

- Now recall that we require the ball to be entirely within the polyhedron. Then the distance of the center of the ball to every hyperplane must be at least r , that is

$$\frac{-a_i^T x + b_i}{\|a_i\|_2} \geq r$$

- Rearranging the above inequality yields the linear constraint

$$a_i^T c + \|a_i\|_2 r \leq b_i$$

- Hence in order to find the center c and the largest possible radius r we need to solve the LP

$$\begin{aligned} \max_{c, r} \quad & r \\ \text{s.t.} \quad & a_i^T c + \|a_i\|_2 r \leq b_i, \quad i = 1, \dots, m \\ & c \in \mathbb{R}^n, \quad r \geq 0 \end{aligned}$$

The optimal solution of this LP gives us the largest inscribed ball within a polyhedron.